Regulatory Status of NAD+ (Nicotinamide Adenine Dinucleotide)

Under the U.S. Food and Drug Administration

Summary prepared August 5, 2026

1. Executive Summary

NAD+ (nicotinamide adenine dinucleotide) has **not been approved by the U.S. Food and Drug Administration (FDA) as a drug** for any medical indication. There is no FDA-approved commercial NAD+ product in oral, injectable, or intravenous form that has completed the New Drug Application (NDA) process demonstrating safety and effectiveness for a specific therapeutic use.

NAD+ remains available through dietary supplements (or precursors) and via pharmacy compounding under defined regulatory pathways, but these pathways do not constitute FDA drug approval.

2. Key Regulatory Distinctions

Understanding the status of NAD+ requires distinguishing between three regulatory categories:

FDA Drug Approval

Applies only to products that have undergone the full NDA process, including clinical trials, manufacturing controls, and approved labeling for a defined medical use. NAD+ has not completed this pathway and is not an FDA-approved drug product.

Pharmacy Compounding (Sections 503A and 503B)

Injectable and intravenous NAD+ preparations are typically compounded by state-licensed pharmacies (Section 503A, patient-specific prescriptions) or, in limited circumstances, by 503B outsourcing facilities. Compounding is permitted under specific statutory conditions but does not equate to FDA approval of the finished product for safety or efficacy. NAD+ has been nominated for inclusion on FDA bulk drug substance lists and has been under evaluation; it is not on the finalized 503B Bulks List in the same manner as certain other substances.

Dietary Supplements

Under the Dietary Supplement Health and Education Act (DSHEA), the FDA does not pre-approve dietary supplements for safety or effectiveness before marketing. Manufacturers bear primary responsibility for product safety and accurate labeling. Products making disease treatment or prevention claims may be regulated as unapproved drugs. Precursors such as nicotinamide riboside (NR) are widely available as dietary supplements. Nicotinamide mononucleotide (NMN) previously faced restrictions tied to investigational drug status; the FDA later affirmed that NMN is not excluded from the dietary supplement definition.

3. FDA Communications on Sterile Compounding

In August 2026, the FDA issued a public reminder to compounders regarding the use of appropriate ingredients for sterile drugs. The agency noted that some compounders have used food-grade nicotinamide adenine dinucleotide (NAD+) sold by repackagers to prepare intravenous products. Food-grade ingredients are not suitable for sterile compounding without appropriate processing because of the elevated risk of microbial and endotoxin contamination.

The FDA has received adverse event reports following use of certain NAD+ injectable products, including severe chills, shaking, vomiting, and fatigue, with some cases requiring medical treatment. These reactions are consistent with excessive levels of endotoxins. The agency urges compounders to verify that bulk ingredients and suppliers are appropriate for drug use and encourages clear labeling by manufacturers and repackagers regarding intended use (food/supplement versus drug).

4. Practical Implications

NAD+ intravenous infusions and injections are commonly offered at wellness and longevity clinics for purported benefits related to energy, recovery, and aging. These remain compounded or off-label offerings. Their safety and effectiveness for these uses have not been established through the FDA drug-approval process. Clinical evidence for direct parenteral NAD+ administration is more limited than data available for certain oral NAD+ precursors that raise NAD+ levels in controlled studies.

Reported side effects associated with infusions can include flushing, nausea, chest tightness, headache, and other transient reactions. Infusion rate adjustments are sometimes used to manage symptoms. Product quality can vary substantially depending on the compounding source, sterility practices, and ingredient grade. Long-term safety data for high-dose injectable NAD+ remain limited.

5. Status Overview

Category	Status
FDA-approved drug product	No — no NDA approval for any indication
Pharmacy compounding (503A)	Permitted under conditions for individual patients; subject to ongoing evaluation
503B outsourcing facilities	Limited; not on the finalized 503B Bulks List for unrestricted use
Dietary supplements / precursors	Available under DSHEA (NR established; NMN affirmed lawful after prior restriction)
Food-grade NAD+ for sterile injectables	Not appropriate; FDA has issued specific warnings

6. Closing Note

NAD+ is a naturally occurring coenzyme essential for cellular energy metabolism, DNA repair, and other biological processes. Related precursors are available as dietary supplements, and compounded preparations may be obtained through licensed pharmacies under applicable rules. However, NAD+ itself has not received FDA approval as a drug. Individuals considering NAD+-related products, particularly injectable or intravenous forms, should consult a qualified healthcare professional, verify the source and quality of any compounded preparation, and remain aware of the current regulatory status and associated risks.

Disclaimer: This document is for informational purposes only and does not constitute medical, legal, or regulatory advice. Regulatory status can change; readers should consult primary FDA sources and qualified professionals for the most current information and for guidance specific to their circumstances. Information is based on publicly available FDA communications, bulk substance evaluations, and regulatory analyses current as of early August 2026.